

SWISS-MODEL — Homology Model Report

Target: BRCA2_HUMAN | UniProt: P51587 | Breast cancer type 2 susceptibility protein
Model 01 | Created: today at 18:20

Model Quality Metrics

Oligo-State	Monomer
GMQE	0.03 (scored over full 3,418-residue submission; only 18% coverage)
QMEANDisCo Global	0.64 ± 0.05 (scored for modelled region only — primary quality indicator)
QSQE	0.00

QMEAN Z-Scores

Deprecated by SWISS-MODEL; included for completeness only.

QMEAN	-4.68
C β	-2.41
All Atom	-1.74
Solvation	-1.41
Torsion	-3.70

Template Information

Template	1mje.1.B — STRUCTURE OF A BRCA2-DSS1-SSDNA COMPLEX
Seq Identity	76.40%
Seq Similarity	0.54
Coverage	0.18
Range	2478–3191
Method	X-ray, 3.50 Å
Biounit Oligo State	Hetero-dimer

Ligand Note

Ligand	DT-DT-DT-DT-DT-DT (oligo-dT ssDNA from template)
Added to Model	No — binding site not conserved
Description	5'-D(P*TP*TP*TP*TP*TP*T)-3'

The template ssDNA was deliberately excluded — leaves a clean receptor pocket for docking.

Interpretation Notes

The low GMQE (0.03) reflects that only 18% of the full 3,418-residue BRCA2 sequence is covered by the template — it is a property of coverage, not model quality. The QMEANDisCo Global score (0.64 ± 0.05) is the relevant quality indicator for the modelled region (residues 2478–3191) and is moderate and trustworthy for a 76.4% identity, 3.5 Å template. QMEAN Z-scores are deprecated by SWISS-MODEL. Local dips in the QMEANDisCo plot should be flagged if best-docking compounds contact those residues.